Experiment 10

Aim: Experimenting with an Advanced NLP Problem using Hugging Face Transformers, spaCy, NLTK (Natural Language Toolkit), AllenNLP, BERTScore for Fine-Tuning Large Language Models (LLMs) for Standard NLP Problems.

Code:

Advanced NLP with Hugging Face Transformers

```
!pip install transformers datasets torch spacy nltk allennlp allennlp-models bert-score
```

```
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  Downloading allennlp_models-2.10.1-py3-none-any.whl.metadata (23 kB) Collecting bert-score
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/usr/local/lib/python3.12/dist-packages (from nltk) (1.5.2)
INFO: pip is looking at multiple versions of allennlp to determine which version is compatible with other requirements. This could take a while.
Collecting allennlp
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  Downloading allennlp-2.9.3-py3-none-any.whl.metadata (19 kB)
  Downloading allennlp-2.9.2-py3-none-any.whl.metadata (19 kB)
  Downloading allennlp-2.9.1-py3-none-any.whl.metadata (19 kB)
  Downloading allennlp-2.9.0-py3-none-any.whl.metadata (18 kB)
  Downloading allennlp-2.8.0-py3-none-any.whl.metadata (17 kB)
  Downloading allennlp-2.7.0-py3-none-any.whl.metadata (17 kB)
INFO: pip is still looking at multiple versions of allennlp to determine which version is compatible with other requirements. This could take a
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  Downloading allennlp-2.5.0-py3-none-any.whl.metadata (17 kB)
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INFO: This is taking longer than usual. You might need to provide the

```
dependency resolver with stricter constraints to reduce runtime. See
https://pip.pypa.io/warnings/backtracking for guidance. If you want to
abort this run, press Ctrl + C.
  Downloading allennlp-2.2.0-py3-none-any.whl.metadata (17 kB)
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Requirement already satisfied: overrides in
/usr/local/lib/python3.12/dist-packages (from allennlp) (7.7.0)
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  Downloading spacy-2.1.9.tar.gz (30.7 MB)
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0:00:00
error: subprocess-exited-with-error

  × pip subprocess to install build dependencies did not run
successfully.
  | exit code: 1
  ╰─> See above for output.

  note: This error originates from a subprocess, and is likely not a
problem with pip.
  Installing build dependencies ... error: subprocess-exited-with-
error

× pip subprocess to install build dependencies did not run
successfully.
| exit code: 1
╰─> See above for output.

note: This error originates from a subprocess, and is likely not a
problem with pip.
```

1. Load Dataset (IMDb Reviews)

```
from datasets import load_dataset

dataset = load_dataset("imdb")
```

```
train_texts = dataset["train"]["text"][:2000]
train_labels = dataset["train"]["label"][:2000]
test_texts = dataset["test"]["text"][:500] test_labels =
dataset["test"]["label"][:500]
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/
_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab
(https://huggingface.co/settings/tokens), set it as secret in your Google Colab
and restart your session.
You will be able to reuse this secret in all of your notebooks. Please note that
authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
```


```
{"model_id":"cfd480ce7da94ad3b638f9f41d3501b0","version_major":2,"vers  ion_minor":0}
{"model_id":"9143e7c5514b4a639a11f12ef5094d9b","version_major":2,"vers  ion_minor":0}
{"model_id":"11b505ca9ade4e2082cbf436fc0e3fec","version_major":2,"vers ion_minor":0}
{"model_id":"7f93c9ddbf4b43f2a9432374ea798c21","version_major":2,"vers  ion_minor":0}
{"model_id":"9b4497d452914c939a3dd0a6a3e4cc73","version_major":2,"vers  ion_minor":0}
{"model_id":"9d1d5fe5f5c54f9d82180b25bec3a5ae","version_major":2,"vers ion_minor":0}
{"model_id":"e40c83d0f953469a9a6db280250971d1","version_major":2,"vers ion_minor":0}
```


2. Hugging Face Transformers (Fine-tune BERT)

```
from transformers import AutoTokenizer,
AutoModelForSequenceClassification, TrainingArguments, Trainer
from datasets import load_dataset
from sklearn.metrics import accuracy_score
import numpy as np
import torch

#  Load dataset
dataset = load_dataset("imdb")
```

```
# Load tokenizer and model
model_name = "distilbert-base-uncased"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForSequenceClassification.from_pretrained(model_name, num_labels=2)

# Tokenization function
def tokenize(batch):
    return tokenizer(batch["text"], padding=True, truncation=True)

# Optional: Use smaller subset for quick demo
dataset_small = {
    split: dataset[split].shuffle(seed=42).select(range(5000)) for split in
    dataset.keys()
}

# Tokenize with batching
encoded_dataset = {}
for split in dataset_small:
    encoded_dataset[split] = dataset_small[split].map( tokenize,
        batched=True,
        batch_size=64, num_proc=4
    )

# Convert to PyTorch format
for split in encoded_dataset: encoded_dataset[split].set_format("torch",
    columns=["input_ids",
"attention_mask", "label"])

# Prepare train/test datasets
train_dataset = encoded_dataset["train"].shuffle(seed=42).select(range(2000))
test_dataset = encoded_dataset["test"].shuffle(seed=42).select(range(500))

# Define metrics for evaluation
def compute_metrics(p):
    preds = np.argmax(p.predictions, axis=1)
    return {"accuracy": accuracy_score(p.label_ids, preds)}

# Training arguments
training_args = TrainingArguments(
    output_dir="./results",
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    num_train_epochs=1, save_strategy="no",
    logging_dir="./logs",
```

```
        report_to=[]
)

# Trainer
trainer = Trainer( model=model,
        args=training_args,
        train_dataset=train_dataset, eval_dataset=test_dataset,
        compute_metrics=compute_metrics
)

# Train the model
trainer.train()

# Evaluate on test set
results = trainer.evaluate()
print("Test Accuracy:", results["eval_accuracy"])
```

```
Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized: ['classifier.bias', 'classifier.weight', 'pre classifier.bias', 'pre classifier.weight']
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
```


```
{"model_id":"e8b5a563f2f049eb97432a36a7ed3813","version_major":2,"vers ion_minor":0}
{"model_id":"28ba607533e64ea7b5b04c6d02cb6cd1","version_major":2,"vers ion_minor":0}
{"model_id":"be12dc6a9b644e5da77ff59fe3974ed2","version_major":2,"vers ion_minor":0}
```


```
<IPython.core.display.HTML object>
<IPython.core.display.HTML object>
Test Accuracy: 0.898
```

```
example_text = "The movie was absolutely fantastic, I loved it!"

# Tokenize the example
inputs = tokenizer(example_text, return_tensors="pt", padding=True, truncation=True)

# Move tensors to the same device as the model
device = torch.device("cuda" if torch.cuda.is_available() else "cpu") model.to(device)
inputs = {k: v.to(device) for k, v in inputs.items()}
```

```
# Get model predictions
with torch.no_grad():
    outputs = model(**inputs) logits =
    outputs.logits
    predicted_class_id = logits.argmax().item()

# Map predicted class to label
label_map = {0: "NEGATIVE", 1: "POSITIVE"}
print("Predicted label:", label_map[predicted_class_id])
```

```
Predicted label: POSITIVE
```

```
example_text = "The movie was absolutely horrible, I hate it!"
# Tokenize the example
inputs = tokenizer(example_text, return_tensors="pt", padding=True, truncation=True)

# Move tensors to the same device as the model
device = torch.device("cuda" if torch.cuda.is_available() else "cpu") model.to(device)
inputs = {k: v.to(device) for k, v in inputs.items()}

# Get model predictions
with torch.no_grad():
    outputs = model(**inputs) logits =
    outputs.logits
    predicted_class_id = logits.argmax().item()

# Map predicted class to label
label_map = {0: "NEGATIVE", 1: "POSITIVE"}
print("Predicted label:", label_map[predicted_class_id])
```

```
Predicted label: NEGATIVE
```

3. spaCy (Pipeline for Sentiment)

```
import spacy
from spacy.training.example import Example
from datasets import load_dataset

#  Load IMDb dataset and take a small subset
dataset = load_dataset("imdb")
train_texts = dataset["train"]["text"][:1000]
train_labels = dataset["train"]["label"][:1000]

# Create blank spaCy model and text classifier
nlp = spacy.blank("en")
```

```
textcat = nlp.add_pipe("textcat")
textcat.add_label("POSITIVE")
textcat.add_label("NEGATIVE")

# Prepare training examples
examples = []
for text, label in zip(train_texts, train_labels):
    cats = {"POSITIVE": bool(label), "NEGATIVE": not bool(label)}
    examples.append(Example.from_dict(nlp.make_doc(text), {"cats":
cats}))

# Initialize optimizer and train
optimizer = nlp.initialize() for
epoch in range(3):
    losses = {}
    nlp.update(examples, sgd=optimizer, losses=losses) print(f"Epoch
    {epoch+1} Losses:", losses)

# Test spaCy model
doc = nlp("This movie was absolutely amazing, I loved it!")
print("spaCy Prediction:", doc.cats)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/
_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab
(https://huggingface.co/settings/tokens), set it as secret in your Google Colab
and restart your session.
You will be able to reuse this secret in all of your notebooks. Please note that
authentication is recommended but still optional to access public models or datasets.
  warnings.warn(

Epoch 1 Losses: {'textcat': 0.25}
Epoch 2 Losses: {'textcat': 0.22656100988388062}
Epoch 3 Losses: {'textcat': 0.19281552731990814}
spaCy Prediction: {'POSITIVE': 0.461971253156662, 'NEGATIVE': 0.5380287766456604}
```

4. NLTK (Preprocessing + Naive Bayes Sentiment)

```
import nltk
from nltk.corpus import movie_reviews
import random

nltk.download("movie_reviews")
```

```
documents = [(list(movie_reviews.words(fileid)), category)
             for category in movie_reviews.categories()
             for fileid in movie_reviews.fileids(category)]
random.shuffle(documents)

all_words = nltk.FreqDist(w.lower() for w in movie_reviews.words())
word_features = list(all_words)[:2000]

def document_features(doc):
    words = set(doc)
    return {w: (w in words) for w in word_features}

featuresets = [(document_features(d), c) for (d, c) in documents]
train_set, test_set = featuresets[100:], featuresets[:100]
classifier = nltk.NaiveBayesClassifier.train(train_set)

print("NLTK Accuracy:", nltk.classify.accuracy(classifier, test_set))
```

```
[nltk_data] Downloading package movie_reviews to /root/nltk_data...
[nltk_data]   Unzipping corpora/movie_reviews.zip.
```

```
NLTK Accuracy: 0.8
```

5. BERT - SCORE Evaluation

```
!pip install bert-score
```

```
Collecting bert-score
  Using cached bert_score-0.3.13-py3-none-any.whl.metadata (15 kB)
Requirement already satisfied: torch>=1.0.0 in
/usr/local/lib/python3.12/dist-packages (from bert-score)
(2.8.0+cu126)
Requirement already satisfied: pandas>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from bert-score) (2.2.2)
Requirement already satisfied: transformers>=3.0.0 in
/usr/local/lib/python3.12/dist-packages (from bert-score) (4.56.1)
Requirement already satisfied: numpy in
/usr/local/lib/python3.12/dist-packages (from bert-score) (2.0.2)
Requirement already satisfied: requests in
/usr/local/lib/python3.12/dist-packages (from bert-score) (2.32.4)
Requirement already satisfied: tqdm>=4.31.1 in
/usr/local/lib/python3.12/dist-packages (from bert-score) (4.67.1)
Requirement already satisfied: matplotlib in
/usr/local/lib/python3.12/dist-packages (from bert-score) (3.10.0)
Requirement already satisfied: packaging>=20.9 in
/usr/local/lib/python3.12/dist-packages (from bert-score) (25.0)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.12/dist-packages (from pandas>=1.0.1->bert-
```

score) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
/usr/local/lib/python3.12/dist-packages (from pandas>=1.0.1->bert-score) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in
/usr/local/lib/python3.12/dist-packages (from pandas>=1.0.1->bert-score) (2025.2)
Requirement already satisfied: filelock in
/usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (3.19.1)
Requirement already satisfied: typing-extensions>=4.10.0 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (4.15.0)
Requirement already satisfied: setuptools in
/usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (1.13.3)
Requirement already satisfied: networkx in
/usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (3.5)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (3.1.6)
Requirement already satisfied: fsspec in
/usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (2025.3.0)
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/usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (12.6.77)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.6.77 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (12.6.77)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.6.80 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (12.6.80)
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (9.10.2.21)
Requirement already satisfied: nvidia-cublas-cu12==12.6.4.1 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (12.6.4.1)
Requirement already satisfied: nvidia-cufft-cu12==11.3.0.4 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (11.3.0.4)
Requirement already satisfied: nvidia-curand-cu12==10.3.7.77 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (10.3.7.77)

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Requirement already satisfied: nvidia-cusolver-cu12==11.7.1.2 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (11.7.1.2)
Requirement already satisfied: nvidia-cusparse-cu12==12.5.4.2 in
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Requirement already satisfied: nvidia-cufile-cu12==1.11.1.6 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (1.11.1.6)
Requirement already satisfied: triton==3.4.0 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (3.4.0)
Requirement already satisfied: huggingface-hub<1.0,>=0.34.0 in
/usr/local/lib/python3.12/dist-packages (from transformers>=3.0.0-
>bert-score) (0.34.4)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.12/dist-packages (from transformers>=3.0.0-
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>bert-score) (2024.11.6)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in
/usr/local/lib/python3.12/dist-packages (from transformers>=3.0.0-
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>bert-score) (0.6.2)
Requirement already satisfied: contourpy>=1.0.1 in
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/usr/local/lib/python3.12/dist-packages (from matplotlib->bert-score) (0.12.1)
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/usr/local/lib/python3.12/dist-packages (from matplotlib->bert-score) (4.59.2)
Requirement already satisfied: kiwisolver>=1.3.1 in
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/usr/local/lib/python3.12/dist-packages (from matplotlib->bert-score) (1.4.9)
Requirement already satisfied: pillow>=8 in
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/usr/local/lib/python3.12/dist-packages (from matplotlib->bert-score) (3.2.3)
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/usr/local/lib/python3.12/dist-packages (from requests->bert-score) (3.4.3)
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/usr/local/lib/python3.12/dist-packages (from requests->bert-score) (3.10)
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Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from requests->bert-score) (2025.8.3)
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already satisfied: six>=1.5 in
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>pandas>=1.0.1->bert-score) (1.17.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3-
>torch>=1.0.0->bert-score) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.0.0-
>bert-score) (3.0.2)
Downloading bert_score-0.3.13-py3-none-any.whl (61 kB)
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```python
from bert_score import score

# Candidate sentences (model outputs or predictions)
cands = [
    "This movie was fantastic!",
    "The plot was boring and predictable.", "I loved
    the cinematography and acting."
]

# Reference sentences (ground truth or expected)
refs = [
    "I really enjoyed this film.",
    "The story was dull and uninteresting.", "The acting
    and visuals were amazing."
```

```
]

# Compute BERTScore
P, R, F1 = score(cands, refs, lang="en")

# Display comparison
for i, (cand, ref) in enumerate(zip(cands, refs)): print(f"Example
    {i+1}:")
    print(f"    Candidate: {cand}")
    print(f"    Reference: {ref}")
    print(f"    BERTScore -> Precision: {P[i]:.4f}, Recall: {R[i]:.4f}, F1:
{F1[i]:.4f}")
    if F1[i] > 0.85:
        print("    Semantically very similar") elif
    F1[i] > 0.65:
        print("    Moderately similar")
    else:
        print("    Semantically different")
    print()
```

```
Some weights of RobertaModel were not initialized from the model checkpoint at
roberta-large and are newly initialized: ['pooler.dense.bias', 'pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able to use it for
predictions and inference.

Example 1:
  Candidate: This movie was fantastic!
  Reference: I really enjoyed this film.
  BERTScore -> Precision: 0.9022, Recall: 0.9247, F1: 0.9133
    Semantically very similar

Example 2:
  Candidate: The plot was boring and predictable.
  Reference: The story was dull and uninteresting.
  BERTScore -> Precision: 0.9622, Recall: 0.9506, F1: 0.9564
    Semantically very similar

Example 3:
  Candidate: I loved the cinematography and acting.
  Reference: The acting and visuals were amazing.
  BERTScore -> Precision: 0.9264, Recall: 0.9303, F1: 0.9284
    Semantically very similar
```